

MongoDB Operations Assignment

Database and Library System

This report demonstrates database setup, insert, read, update, query-operator, and delete operations in MongoDB. Each command is followed by its corresponding mongosh screenshot.

Student Database

1. Database & Collection Setup:

1. Create a new database

Command

```
use StudentDB
```

2. Create a collection

The student collection is created automatically when the first `db.student.insertOne()` command is executed.

2. Insert Operations

1. Insert One Document

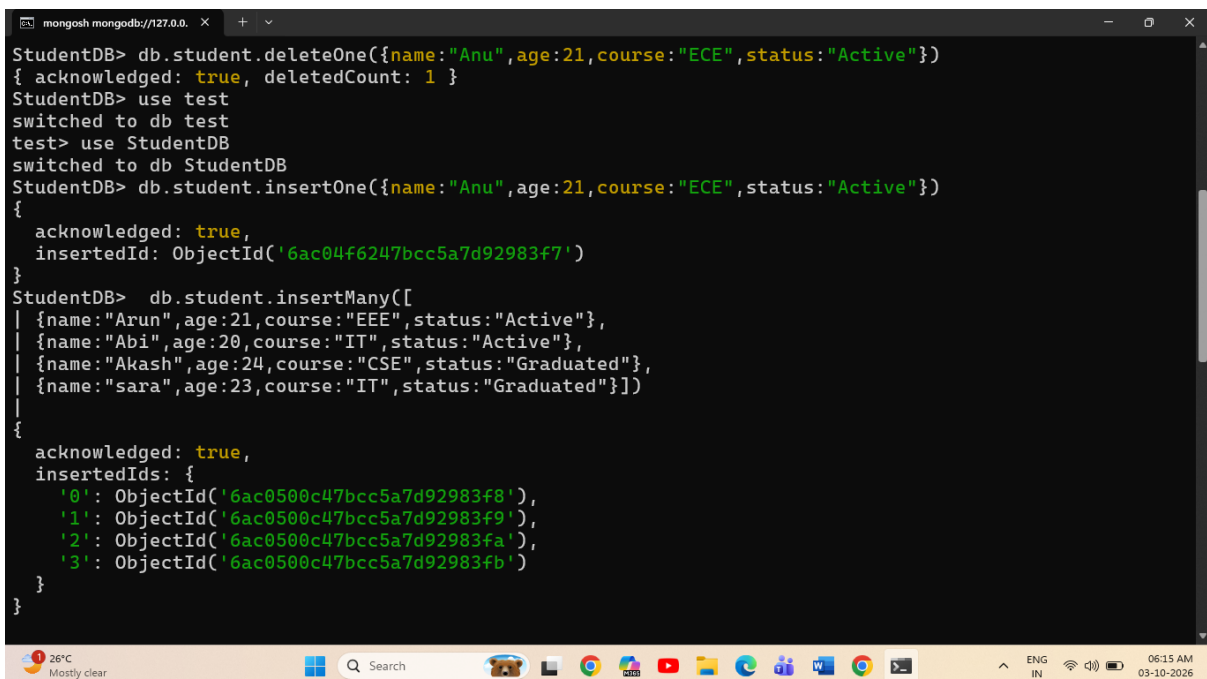
Command

```
db.student.insertOne({name:"Anu",age:21,course:"ECE",status:"Active"})
```

2. Insert Multiple Documents

Command

```
db.student.insertMany([
  {name:"Arun",age:21,course:"EEE",status:"Active"},
  {name:"Abi",age:20,course:"IT",status:"Active"},
  {name:"Akash",age:24,course:"CSE",status:"Graduated"},
  {name:"sara",age:23,course:"IT",status:"Graduated"}
])
```



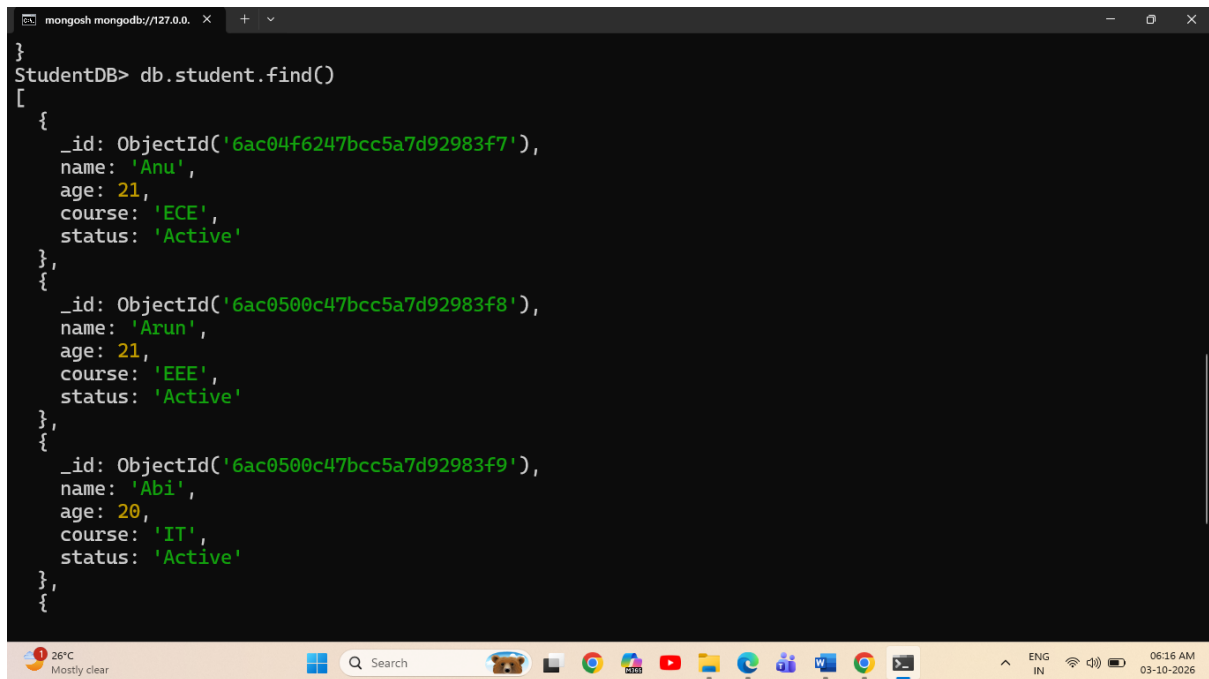
```
mongosh mongodb://127.0.0.1:27017
StudentDB> db.student.deleteOne({name:"Anu",age:21,course:"ECE",status:"Active"})
{ acknowledged: true, deletedCount: 1 }
StudentDB> use test
switched to db test
test> use StudentDB
switched to db StudentDB
StudentDB> db.student.insertOne({name:"Anu",age:21,course:"ECE",status:"Active"})
{
  acknowledged: true,
  insertedId: ObjectId('6ac04f6247bcc5a7d92983f7')
}
StudentDB> db.student.insertMany([
  {name:"Arun",age:21,course:"EEE",status:"Active"},
  {name:"Abi",age:20,course:"IT",status:"Active"},
  {name:"Akash",age:24,course:"CSE",status:"Graduated"},
  {name:"sara",age:23,course:"IT",status:"Graduated"}
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac0500c47bcc5a7d92983f8'),
    '1': ObjectId('6ac0500c47bcc5a7d92983f9'),
    '2': ObjectId('6ac0500c47bcc5a7d92983fa'),
    '3': ObjectId('6ac0500c47bcc5a7d92983fb')
  }
}
```

3. Read Operations

1. Fetch All Documents

Command

`db.student.find()`

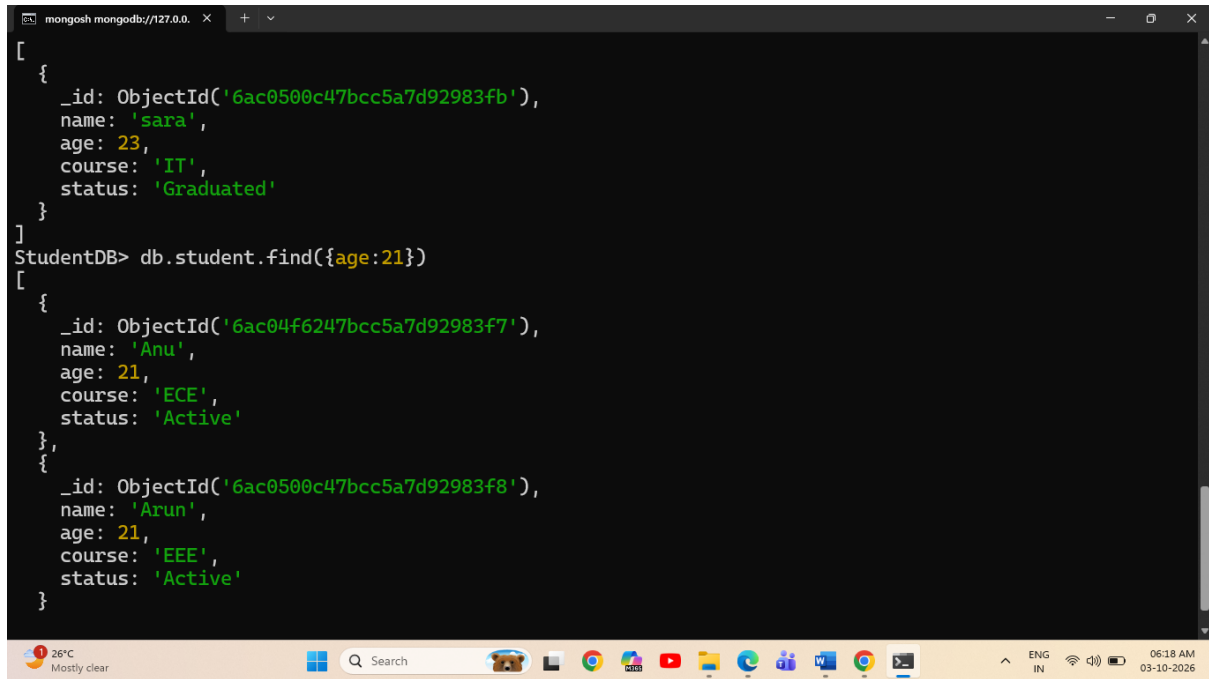


```
}
StudentDB> db.student.find()
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f9'),
    name: 'Abi',
    age: 20,
    course: 'IT',
    status: 'Active'
  },
]
```

2.Filter Students by Age

Command

```
db.student.find({age:21})
```



The screenshot shows a terminal window titled 'mongosh mongodb://127.0.0.1'. It displays a JSON document for a student named 'sara' with age 23, course 'IT', and status 'Graduated'. Below this, the command 'StudentDB> db.student.find({age:21})' is entered. The output shows two documents: one for 'Anu' (age 21, course 'ECE', status 'Active') and one for 'Arun' (age 21, course 'EEE', status 'Active'). The Windows taskbar at the bottom shows the date as 03-10-2026 and time as 06:18 AM.

```
[
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fb'),
    name: 'sara',
    age: 23,
    course: 'IT',
    status: 'Graduated'
  }
]
StudentDB> db.student.find({age:21})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Active'
  }
]
```

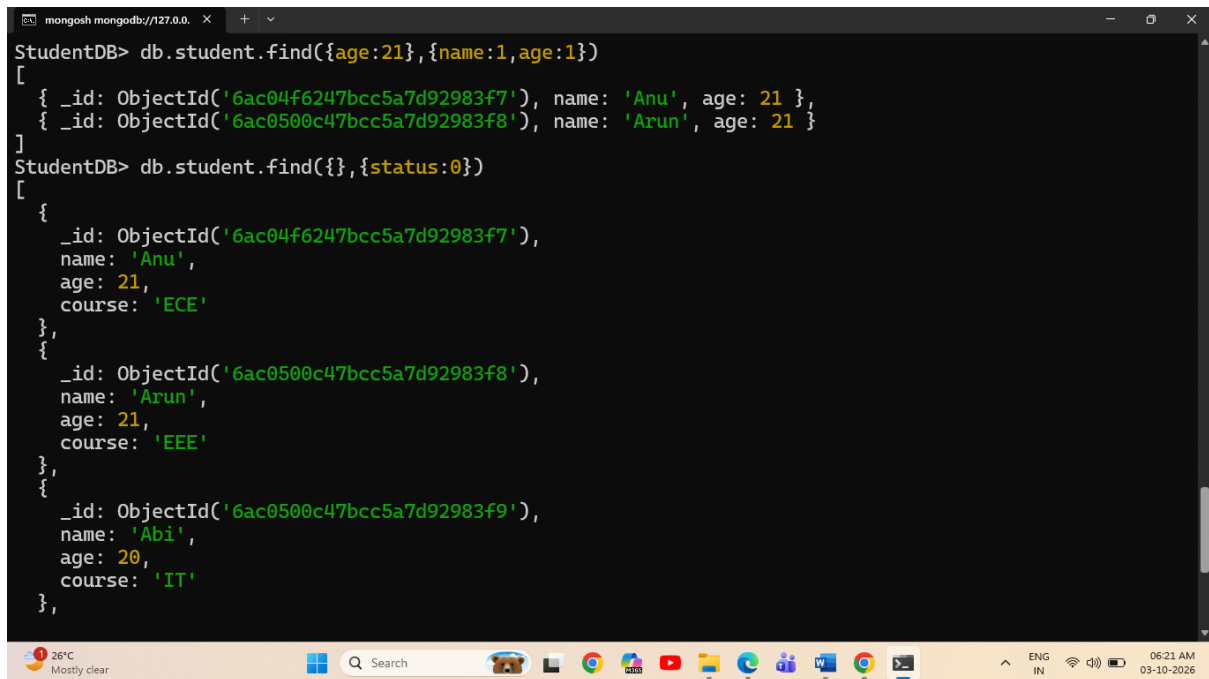
3. Project and Exclude Fields

Command(included)

```
db.student.find({age:21},{name:1,age:1})
```

Exclusion command

```
db.student.find({}, {status:0})
```



The screenshot shows a MongoDB command prompt window with the following content:

```
mongosh mongodb://127.0.0.1:27020
StudentDB> db.student.find({age:21},{name:1,age:1})
[
  { _id: ObjectId('6ac04f6247bcc5a7d92983f7'), name: 'Anu', age: 21 },
  { _id: ObjectId('6ac0500c47bcc5a7d92983f8'), name: 'Arun', age: 21 }
]
StudentDB> db.student.find({}, {status:0})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f9'),
    name: 'Abi',
    age: 20,
    course: 'IT'
  },
]
```

The window also shows a Windows taskbar at the bottom with the date and time 06:21 AM 03-10-2026.

4. Update Operations

1. Update One Document

Command

```
db.student.updateOne({status:"Active"},{$set:{status:"Graduated"}})
```



```
mongosh mongodb://127.0.0.1:27020
StudentDB> db.student.updateOne({status:"Active"},{$set:{status:"Graduated"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
StudentDB> db.student.find()
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Active'
  }
]
```

The screenshot shows a terminal window with the MongoDB command prompt. The first command updates the status of the document with status 'Active' to 'Graduated'. The second command lists all documents in the 'student' collection, showing that the first document (Anu) now has a status of 'Graduated' and the second document (Arun) remains 'Active'.

2. Update Multiple Documents

Command

```
db.student.updateMany({age:21},{ $set:{status:"Graduated"}})
```



```
mongosh mongodb://127.0.0.1 x + v
StudentDB> db.student.updateMany({age:21},{ $set:{status:"Graduated"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 1,
  upsertedCount: 0
}
StudentDB> db.student.find()
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Graduated'
  },
  {
  }
```

The screenshot shows a terminal window with the MongoDB command prompt. The first command executed is `db.student.updateMany({age:21},{ $set:{status:"Graduated"}})`, which returns a status object indicating that 2 documents were matched, 1 was modified, and 0 were upserted. The second command is `db.student.find()`, which returns a list of two documents. Both documents have an age of 21 and a status of 'Graduated'. The first document is for 'Anu' in the 'ECE' course, and the second is for 'Arun' in the 'EEE' course. The terminal window has a title bar that says 'mongosh mongodb://127.0.0.1 x + v'. The bottom of the screen shows a Windows taskbar with various icons and a system tray showing the date and time as 06:29 AM on 03-10-2026.

5 Query Operators

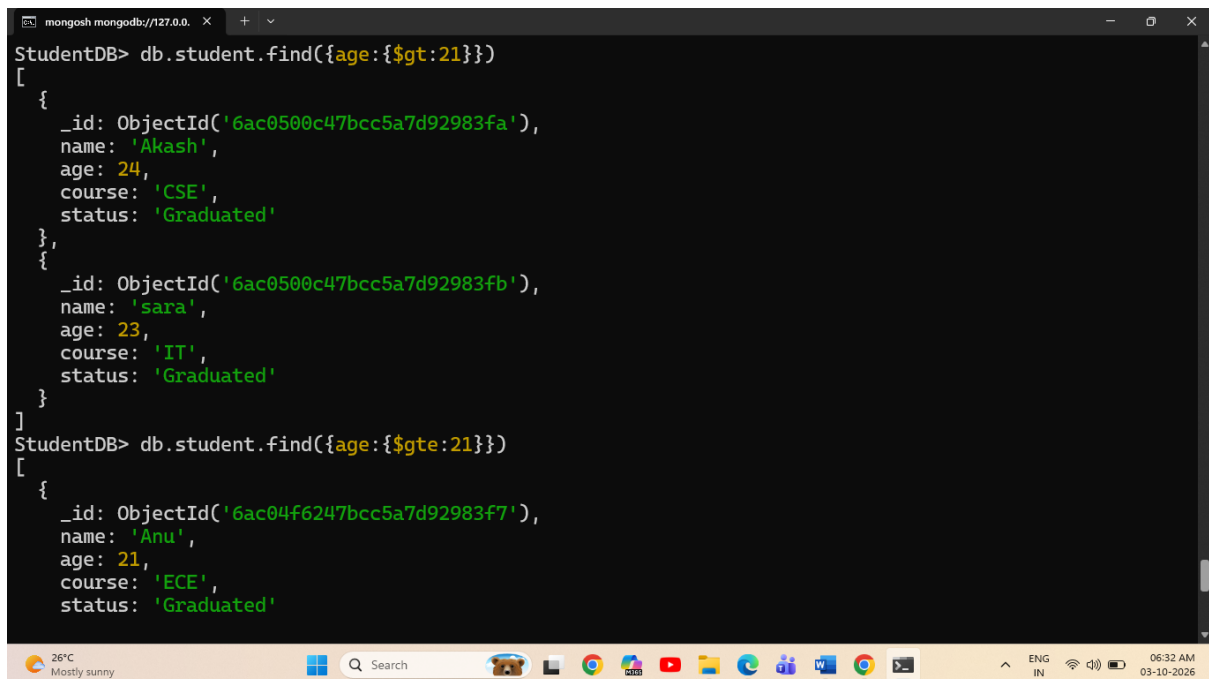
Greater Than and Greater Than or Equal To

\$gt command

```
db.student.find({age:{$gt:21}})
```

\$gte command

```
db.student.find({age:{$gte:21}})
```



The screenshot shows a MongoDB command prompt window with the following content:

```
mongosh mongodb://127.0.0.1:27017
StudentDB> db.student.find({age:{$gt:21}})
[
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fa'),
    name: 'Akash',
    age: 24,
    course: 'CSE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fb'),
    name: 'sara',
    age: 23,
    course: 'IT',
    status: 'Graduated'
  }
]
StudentDB> db.student.find({age:{$gte:21}})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  }
]
```

The window title is "mongosh mongodb://127.0.0.1:27017". The bottom status bar shows "26°C Mostly sunny", a search bar, and system icons for Windows, Google, and others, along with the time "06:32 AM 03-10-2026".

Less Than and Less Than or Equal To

\$lt command

```
db.student.find({age:{$lt:23}})
```

\$lte command

```
db.student.find({age:{$lte:23}})
```

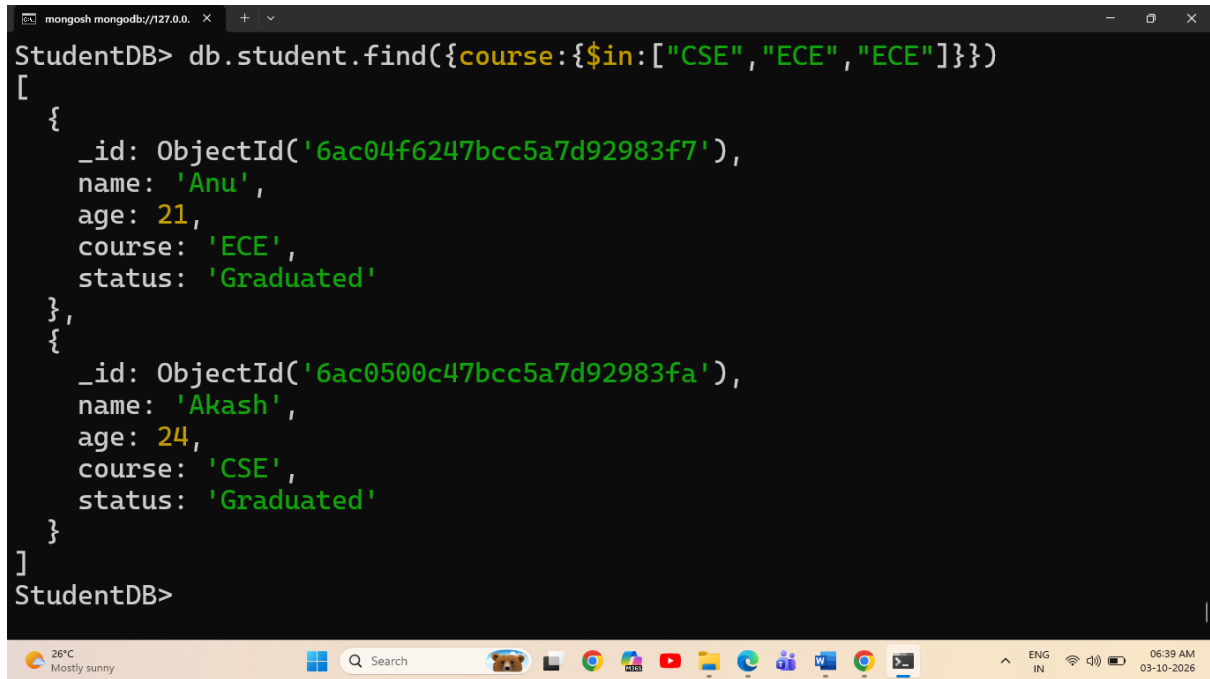
```
mongosh mongodb://127.0.0.1:27017
StudentDB> db.student.find({age:{$lt:23}})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f9'),
    name: 'Abi',
    age: 20,
    course: 'IT',
    status: 'Active'
  }
]
StudentDB> db.student.find({age:{$lte:23}})
```

```
mongosh mongodb://127.0.0.1:27017
StudentDB> db.student.find({age:{$lte:23}})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f9'),
    name: 'Abi',
    age: 20,
    course: 'IT',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fb'),
    name: '...',
    age: 20,
    course: '...',
    status: '...'
  }
]
```

In Operator

Command

```
db.student.find({course:{$in:["CSE","ECE"]}})
```



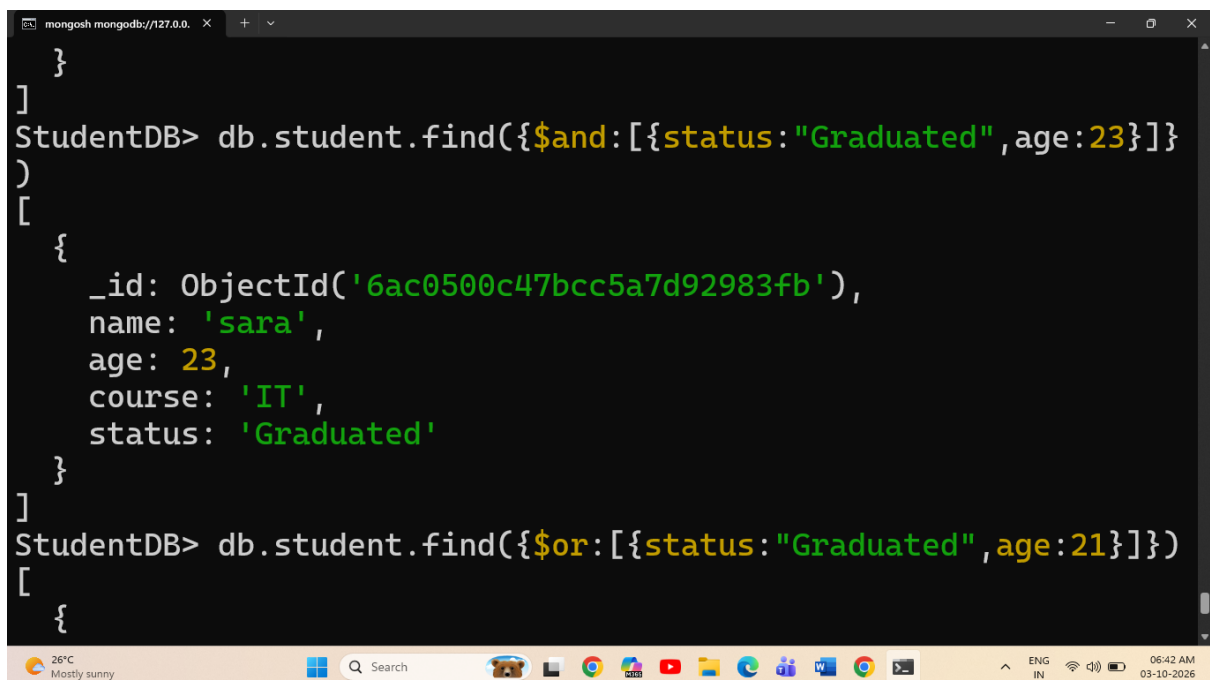
The screenshot shows a MongoDB shell window with the command `db.student.find({course:{$in:["CSE","ECE"]}})` and its output. The output is a JSON array containing two student documents. The first document is for 'Anu', age 21, course 'ECE', and status 'Graduated'. The second document is for 'Akash', age 24, course 'CSE', and status 'Graduated'.

```
StudentDB> db.student.find({course:{$in:["CSE","ECE"]}})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fa'),
    name: 'Akash',
    age: 24,
    course: 'CSE',
    status: 'Graduated'
  }
]
StudentDB>
```

AND and OR Operators

\$and command

```
db.student.find({$and:[{status:"Graduated"},{age:23}]})
```

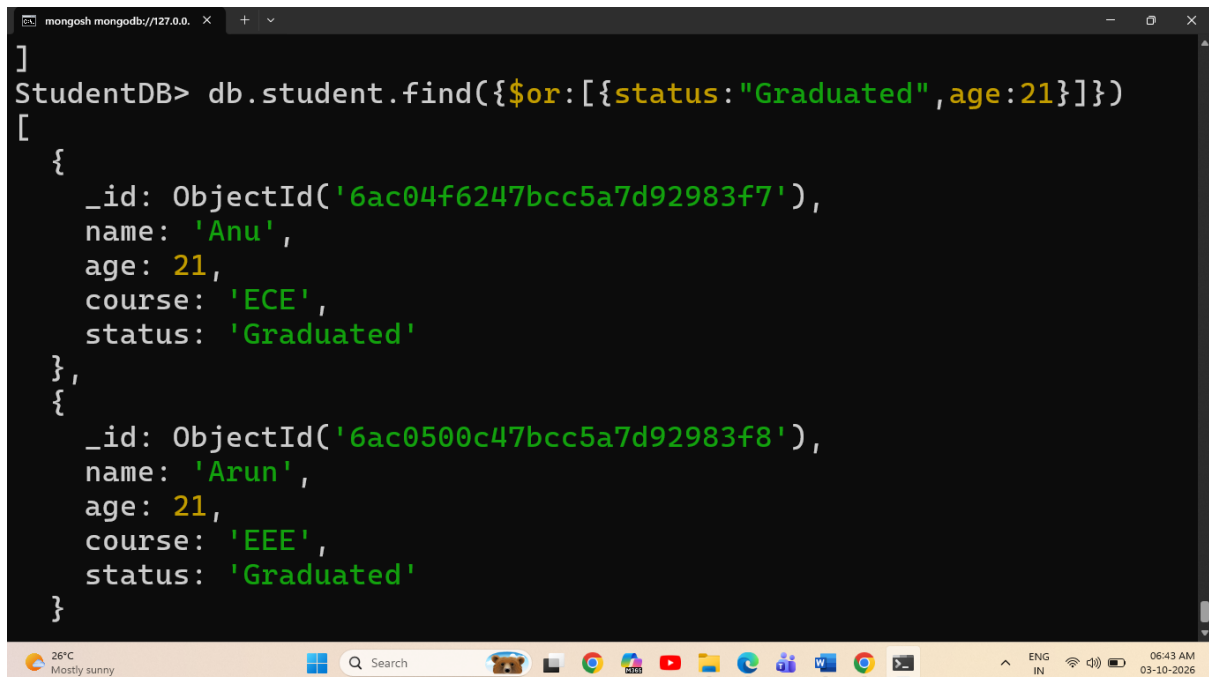


The screenshot shows a MongoDB shell window with the command `db.student.find({$and:[{status:"Graduated"},{age:23}]})` and its output. The output is a JSON array containing one student document for 'sara', age 23, course 'IT', and status 'Graduated'. Below the first query, the command `db.student.find({$or:[{status:"Graduated"},{age:21}]})` is also shown, but its output is not visible in the screenshot.

```
StudentDB> db.student.find({$and:[{status:"Graduated"},{age:23}]})
[
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fb'),
    name: 'sara',
    age: 23,
    course: 'IT',
    status: 'Graduated'
  }
]
StudentDB> db.student.find({$or:[{status:"Graduated"},{age:21}]})
[
  {
```

\$or command

```
db.student.find({$or:[{status:"Graduated"},{age:21}]})
```



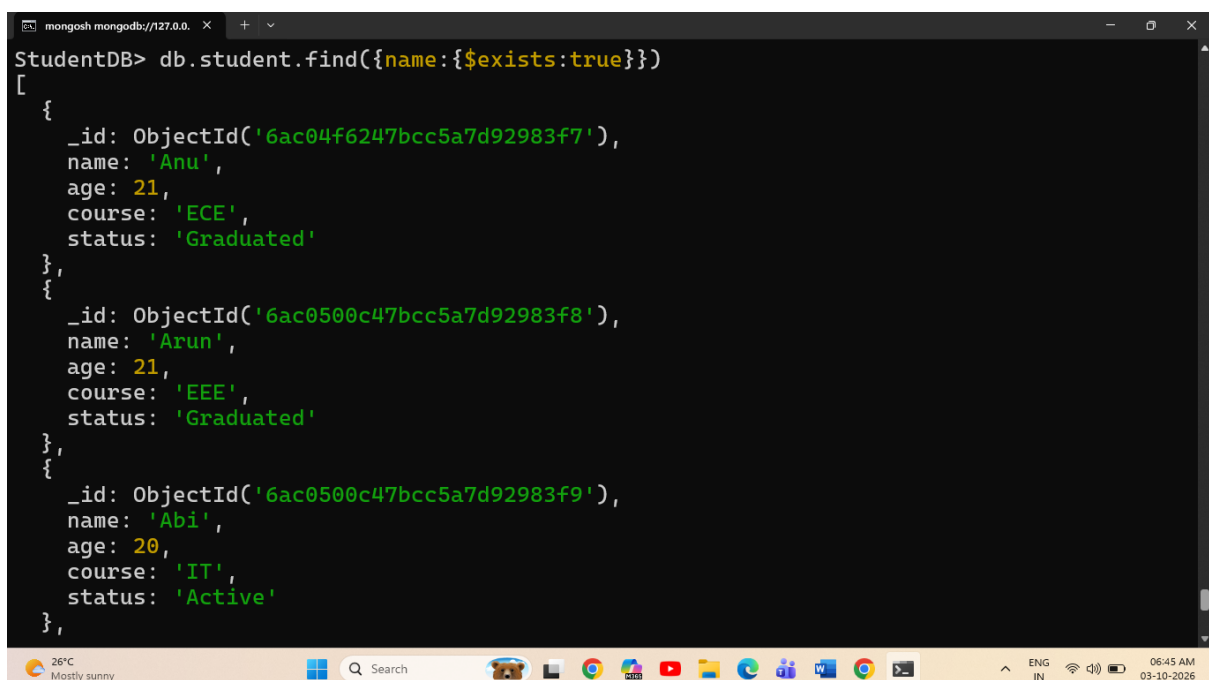
The screenshot shows a MongoDB shell window with the command `db.student.find({$or:[{status:"Graduated"},{age:21}]})` executed. The output is a JSON array containing two documents. The first document is for a student named 'Anu' with age 21, course 'ECE', and status 'Graduated'. The second document is for a student named 'Arun' with age 21, course 'EEE', and status 'Graduated'.

```
StudentDB> db.student.find({$or:[{status:"Graduated"},{age:21}]})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Graduated'
  }
]
```

Exists Operator

Command

```
db.student.find({name:{$exists:true}})
```



The screenshot shows a MongoDB shell window with the command `db.student.find({name:{$exists:true}})` executed. The output is a JSON array containing three documents. The first document is for a student named 'Anu' with age 21, course 'ECE', and status 'Graduated'. The second document is for a student named 'Arun' with age 21, course 'EEE', and status 'Graduated'. The third document is for a student named 'Abi' with age 20, course 'IT', and status 'Active'.

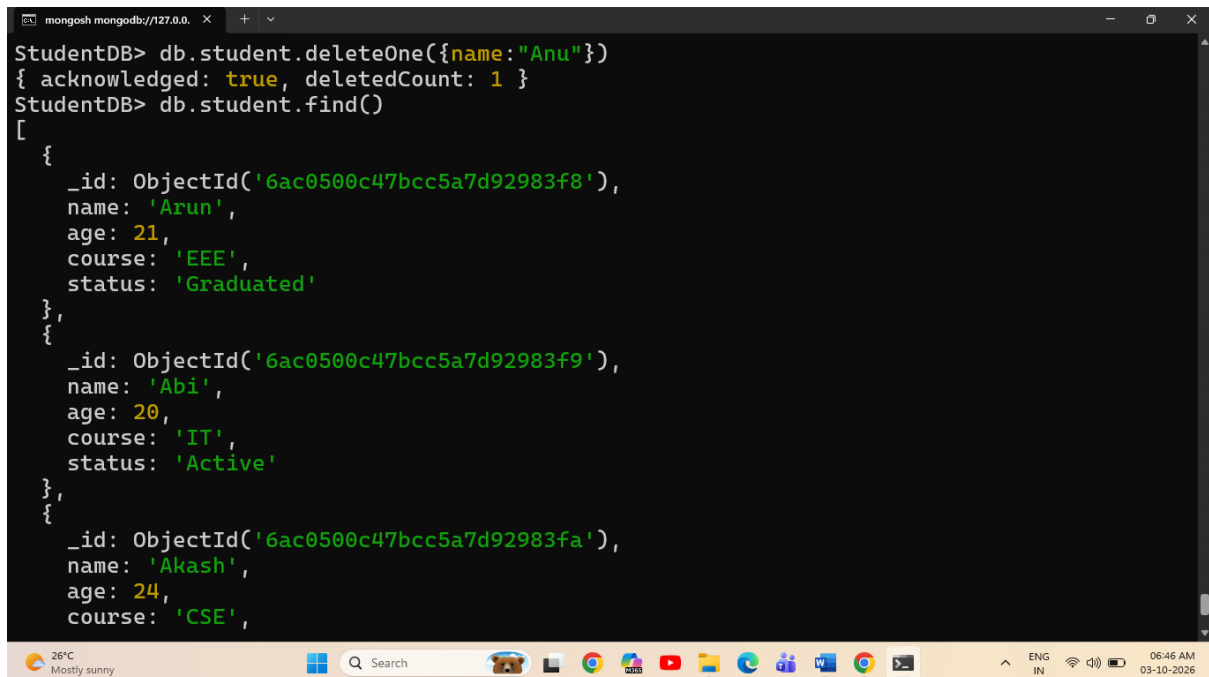
```
StudentDB> db.student.find({name:{$exists:true}})
[
  {
    _id: ObjectId('6ac04f6247bcc5a7d92983f7'),
    name: 'Anu',
    age: 21,
    course: 'ECE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f9'),
    name: 'Abi',
    age: 20,
    course: 'IT',
    status: 'Active'
  }
]
```

6. Delete Operations

Delete One Document

Command

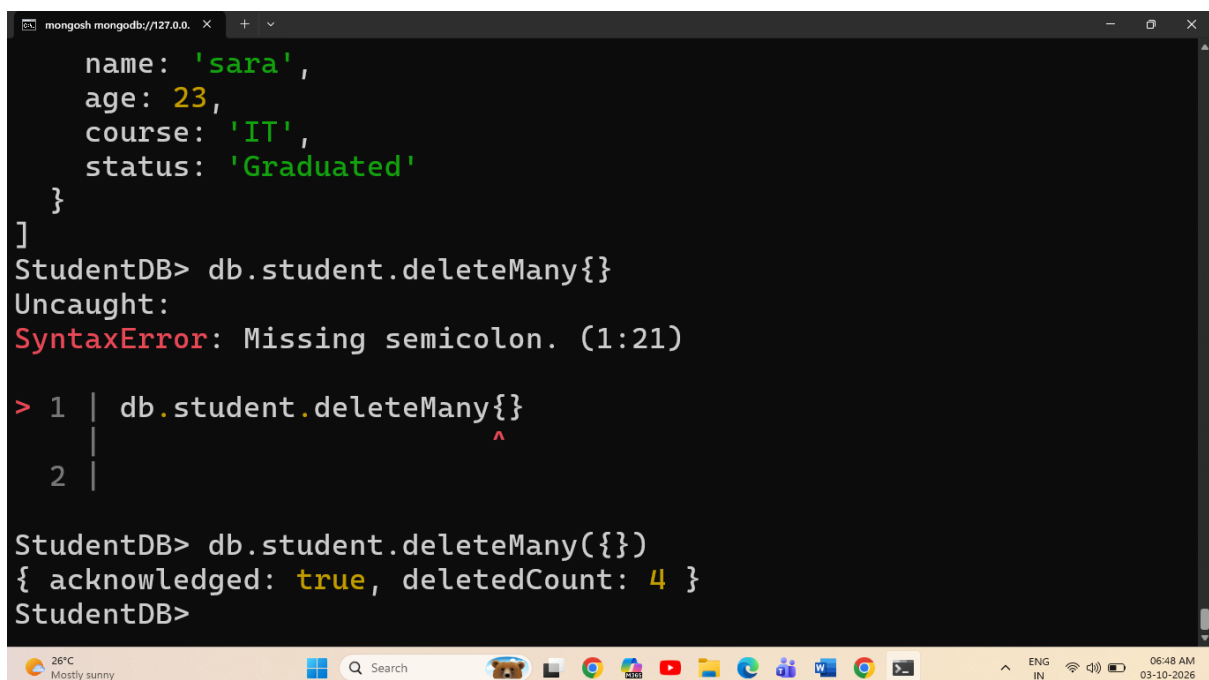
```
db.student.deleteOne({name:"Anu"})
```



```
mongosh mongodb://127.0.0.1:27020
StudentDB> db.student.deleteOne({name:"Anu"})
{ acknowledged: true, deletedCount: 1 }
StudentDB> db.student.find()
[
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f8'),
    name: 'Arun',
    age: 21,
    course: 'EEE',
    status: 'Graduated'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983f9'),
    name: 'Abi',
    age: 20,
    course: 'IT',
    status: 'Active'
  },
  {
    _id: ObjectId('6ac0500c47bcc5a7d92983fa'),
    name: 'Akash',
    age: 24,
    course: 'CSE',
    status: 'Active'
  }
]
```

Delete All Documents

Command : `db.student.deleteMany({})`



```
mongosh mongodb://127.0.0.1:27020
name: 'sara',
age: 23,
course: 'IT',
status: 'Graduated'
}
]
StudentDB> db.student.deleteMany{}
Uncaught:
SyntaxError: Missing semicolon. (1:21)

> 1 | db.student.deleteMany{}
    | ^
    2 |

StudentDB> db.student.deleteMany({})
{ acknowledged: true, deletedCount: 4 }
StudentDB>
```

2 Library System

1. Database Setup and Insert Operations

Database command

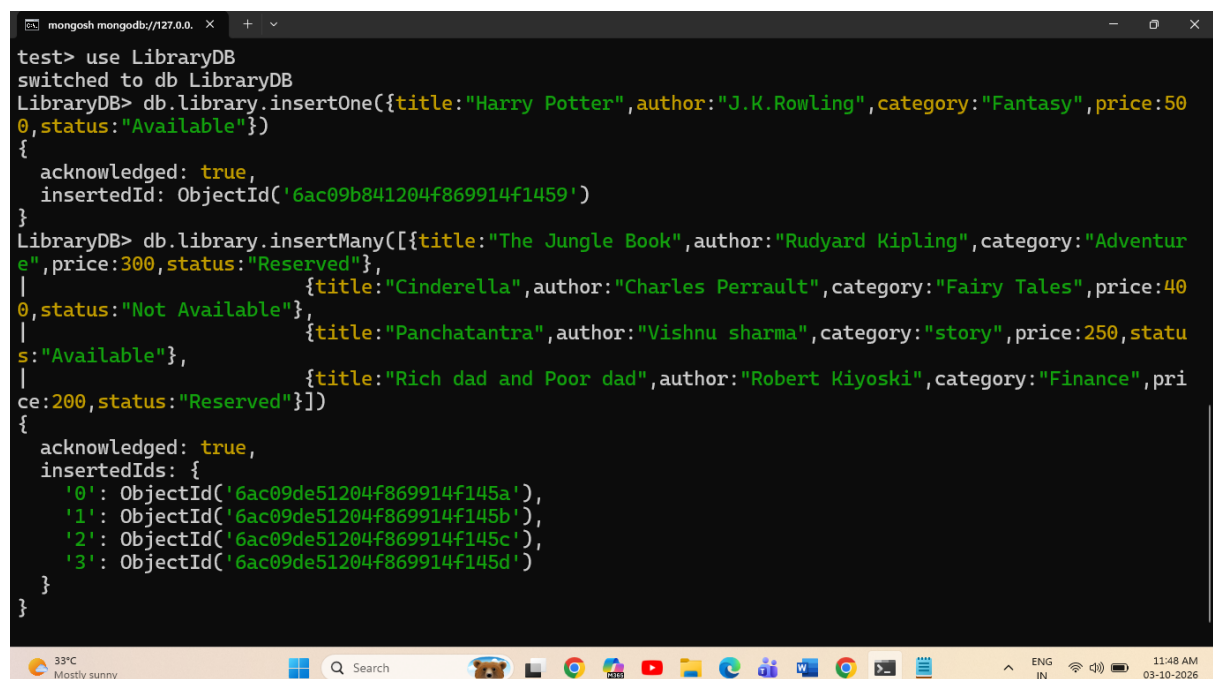
use LibraryDB

Insert one command

```
db.library.insertOne({title:"Harry  
Potter",author:"J.K.Rowling",category:"Fantasy",price:500,status:"Available"})
```

Insert many command

```
db.library.insertMany([  
{title:"The Jungle Book",author:"Rudyard  
Kipling",category:"Adventure",price:300,status:"Reserved"},  
{title:"Cinderella",author:"Charles Perrault",category:"Fairy  
Tales",price:400,status:"Not Available"},  
{title:"Panchatantra",author:"Vishnu  
sharma",category:"story",price:250,status:"Available"},  
{title:"Rich dad and Poor dad",author:"Robert  
Kiyoski",category:"Finance",price:200,status:"Reserved"}  
])
```

A screenshot of a MongoDB terminal window. The window title is 'mongosh mongodbg://127.0.0.1'. The terminal shows the following commands and output:

```
test> use LibraryDB  
switched to db LibraryDB  
LibraryDB> db.library.insertOne({title:"Harry Potter",author:"J.K.Rowling",category:"Fantasy",price:500,status:"Available"})  
{  
  acknowledged: true,  
  insertedId: ObjectId('6ac09b841204f869914f1459')  
}  
LibraryDB> db.library.insertMany([  
  {title:"The Jungle Book",author:"Rudyard Kipling",category:"Adventure",price:300,status:"Reserved"},  
  {title:"Cinderella",author:"Charles Perrault",category:"Fairy Tales",price:400,status:"Not Available"},  
  {title:"Panchatantra",author:"Vishnu sharma",category:"story",price:250,status:"Available"},  
  {title:"Rich dad and Poor dad",author:"Robert Kiyoski",category:"Finance",price:200,status:"Reserved"}  
)  
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6ac09de51204f869914f145a'),  
    '1': ObjectId('6ac09de51204f869914f145b'),  
    '2': ObjectId('6ac09de51204f869914f145c'),  
    '3': ObjectId('6ac09de51204f869914f145d')  
  }  
}
```


The bottom of the screenshot shows a Windows taskbar with various icons, including the Start button, Search, and several application icons. The system tray on the right shows the date and time as '11:48 AM 03-10-2026'.

2.Read All Library Documents

Command

db.library.find()

```
mongosh mongodb://127.0.0.1:27017/
LibraryDB> db.library.find()
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    category: 'Fantasy',
    price: 500,
    status: 'Available'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    category: 'Adventure',
    price: 300,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145b'),
    title: 'Cinderella',
    author: 'Charles Perrault',
    category: 'Fairy Tales',
    price: 400,
    status: 'Not Available'
  },
]
```

```
mongosh mongodb://127.0.0.1:27017/
status: 'Reserved'
},
{
  _id: ObjectId('6ac09de51204f869914f145b'),
  title: 'Cinderella',
  author: 'Charles Perrault',
  category: 'Fairy Tales',
  price: 400,
  status: 'Not Available'
},
{
  _id: ObjectId('6ac09de51204f869914f145c'),
  title: 'Panchatantra',
  author: 'Vishnu sharma',
  category: 'story',
  price: 250,
  status: 'Available'
},
{
  _id: ObjectId('6ac09de51204f869914f145d'),
  title: 'Rich dad and Poor dad',
  author: 'Robert Kiyoski',
  category: 'Finance',
  price: 200,
  status: 'Reserved'
}
```

3. Update Operations

Update One Document

Command

```
db.library.updateOne({status:"Not Available"},{$set:{status:"Available"}})
```

```
mongosh mongodb://127.0.0.1:27017/
LibraryDB> db.library.updateOne({status:"Not Available"},{$set:{status:"Available"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
LibraryDB> db.library.find()
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    category: 'Fantasy',
    price: 500,
    status: 'Available'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    category: 'Adventure',
    price: 300,
    status: 'Reserved'
  },
]
```

```
mongosh mongodb://127.0.0.1:27017/
{
  _id: ObjectId('6ac09de51204f869914f145b'),
  title: 'Cinderella',
  author: 'Charles Perrault',
  category: 'Fairy Tales',
  price: 400,
  status: 'Available'
},
{
  _id: ObjectId('6ac09de51204f869914f145c'),
  title: 'Panchatantra',
  author: 'Vishnu sharma',
  category: 'story',
  price: 250,
  status: 'Available'
},
{
  _id: ObjectId('6ac09de51204f869914f145d'),
  title: 'Rich dad and Poor dad',
  author: 'Robert Kiyoski',
  category: 'Finance',
  price: 200,
  status: 'Reserved'
}
]
LibraryDB>
```

4.Update Multiple Documents

Command

```
db.library.updateMany({status:"Available"},{$set:{status:"Reserved"}})
```

```
mongosh mongodb://127.0.0.1
LibraryDB> db.library.updateMany({status:"Available"},{$set:{status:"Reserved"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
LibraryDB> db.library.find()
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    category: 'Fantasy',
    price: 500,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    category: 'Adventure',
    price: 300,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145b'),
    title: 'Cinderella',
    author: 'Charles Perrault',
    category: 'Fairy Tales',
    price: 400,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145c'),
    title: 'Panchatantra',
    author: 'Vishnu sharma',
    category: 'story',
    price: 250,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145d'),
    title: 'Rich dad and Poor dad',
    author: 'Robert Kiyoski',
    category: 'Finance',
    price: 200,
    status: 'Reserved'
  }
]
LibraryDB>
```

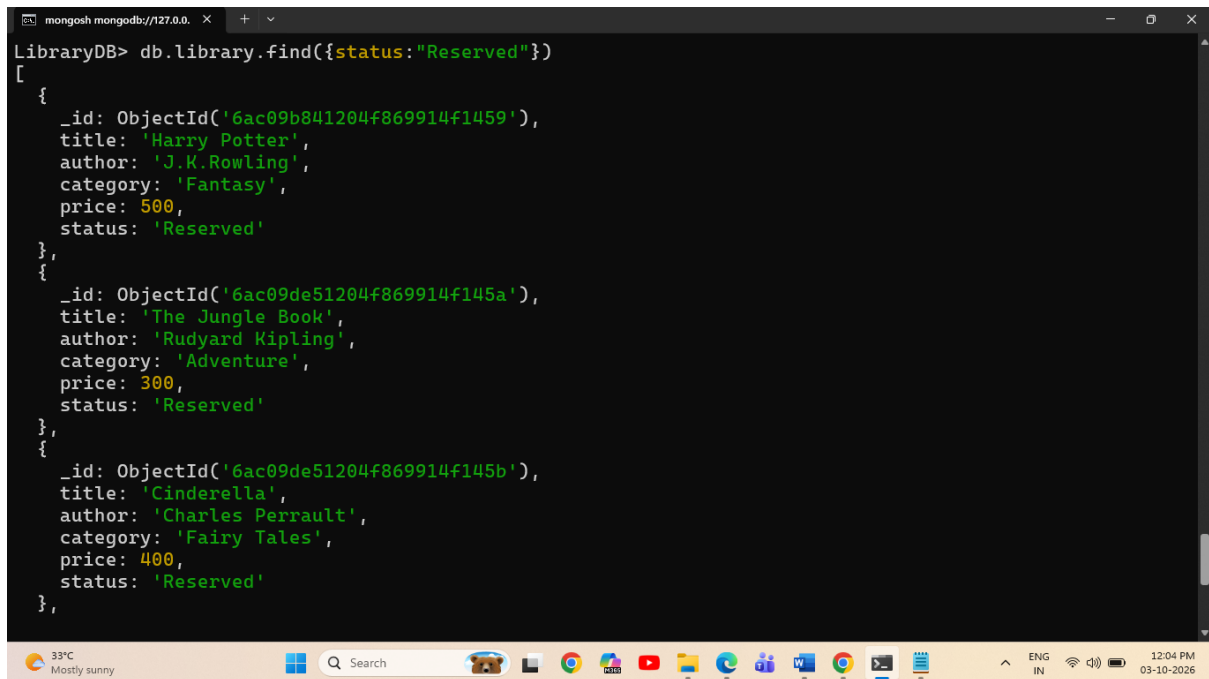
```
mongosh mongodb://127.0.0.1
LibraryDB> db.library.updateMany({status:"Available"},{$set:{status:"Reserved"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
LibraryDB> db.library.find()
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    category: 'Fantasy',
    price: 500,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    category: 'Adventure',
    price: 300,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145b'),
    title: 'Cinderella',
    author: 'Charles Perrault',
    category: 'Fairy Tales',
    price: 400,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145c'),
    title: 'Panchatantra',
    author: 'Vishnu sharma',
    category: 'story',
    price: 250,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145d'),
    title: 'Rich dad and Poor dad',
    author: 'Robert Kiyoski',
    category: 'Finance',
    price: 200,
    status: 'Reserved'
  }
]
LibraryDB>
```

5. Search Operations

1. Search by Status

Command

```
db.library.find({status:"Reserved"})
```



The screenshot shows a MongoDB command prompt window titled 'mongosh mongodb://127.0.0.1'. The command entered is `LibraryDB> db.library.find({status:"Reserved"})`. The output displays three JSON documents representing books with the status 'Reserved':

```
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    category: 'Fantasy',
    price: 500,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    category: 'Adventure',
    price: 300,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145b'),
    title: 'Cinderella',
    author: 'Charles Perrault',
    category: 'Fairy Tales',
    price: 400,
    status: 'Reserved'
  },
]
```

The bottom of the image shows a Windows taskbar with the date and time '12:04 PM 03-10-2026' and system status icons.

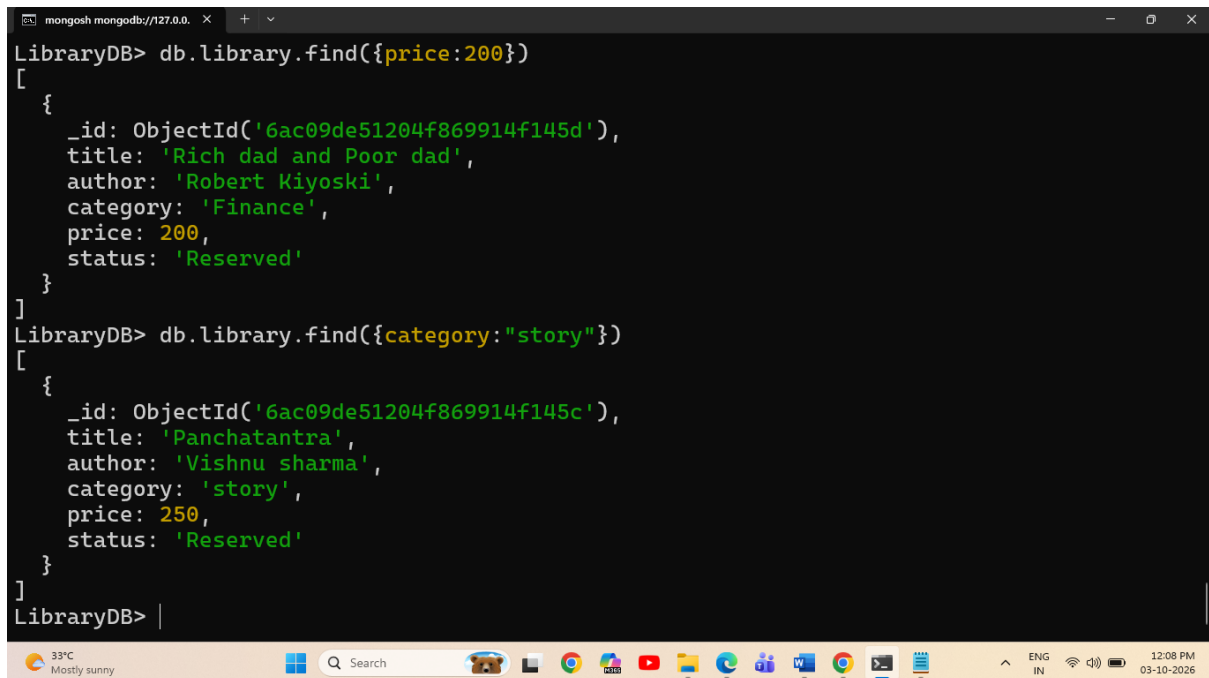
2.Search by Price and Category

Search by price

```
db.library.find({price:200})
```

Search by category

```
db.library.find({category:"story"})
```



The screenshot shows a terminal window titled 'mongosh mongodb://127.0.0.1'. The user is in the 'LibraryDB' database. The first command executed is `db.library.find({price:200})`, which returns a single document for a book titled 'Rich dad and Poor dad' by Robert Kiyoski, categorized as 'Finance' with a price of 200. The second command is `db.library.find({category:"story"})`, which returns a single document for a book titled 'Panchatantra' by Vishnu sharma, categorized as 'story' with a price of 250. The terminal output is as follows:

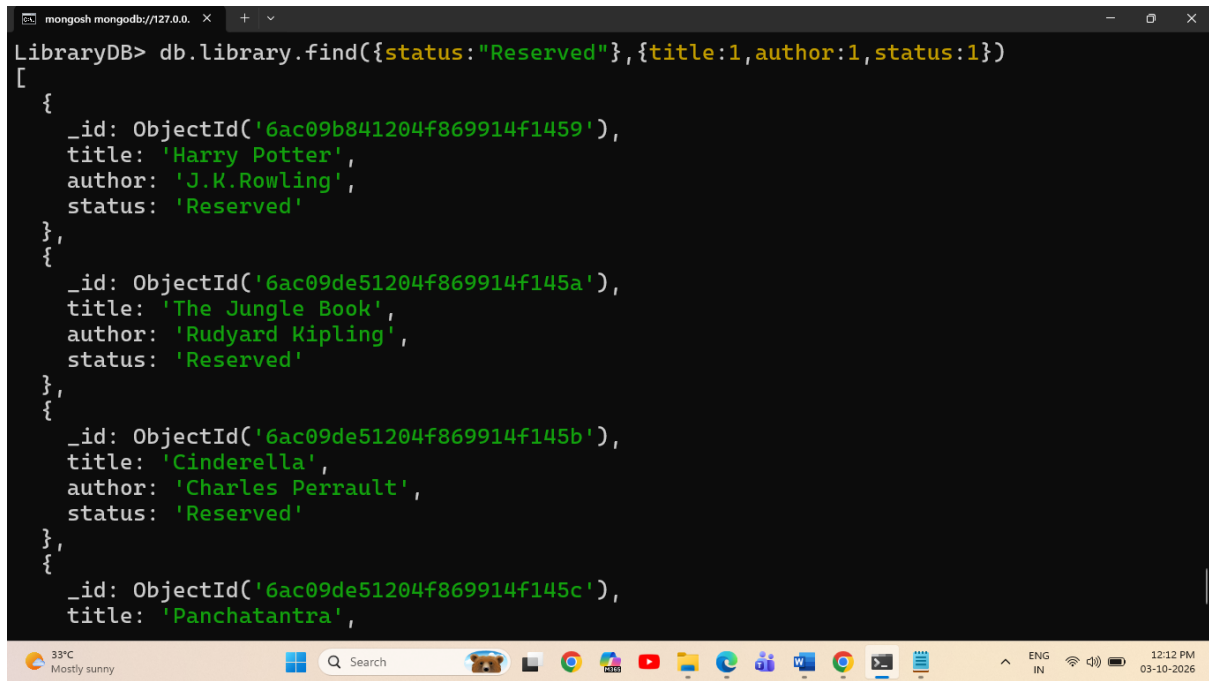
```
LibraryDB> db.library.find({price:200})
[
  {
    _id: ObjectId('6ac09de51204f869914f145d'),
    title: 'Rich dad and Poor dad',
    author: 'Robert Kiyoski',
    category: 'Finance',
    price: 200,
    status: 'Reserved'
  }
]
LibraryDB> db.library.find({category:"story"})
[
  {
    _id: ObjectId('6ac09de51204f869914f145c'),
    title: 'Panchatantra',
    author: 'Vishnu sharma',
    category: 'story',
    price: 250,
    status: 'Reserved'
  }
]
LibraryDB> |
```

The Windows taskbar at the bottom shows the system clock as 12:08 PM on 03-10-2026, with a temperature of 33°C and 'Mostly sunny' weather.

3.Fetch Selected Fields

Command

```
db.library.find({status:"Reserved"},{title:1,author:1,status:1})
```



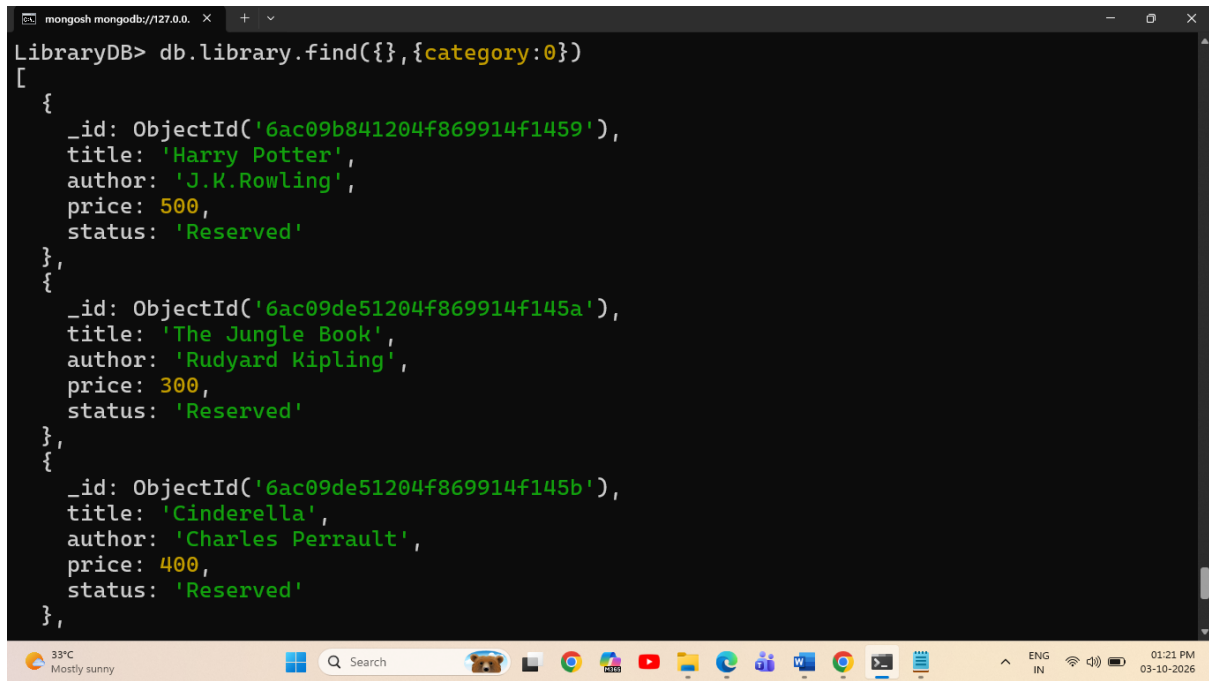
The screenshot shows a terminal window titled 'mongosh mongodb://127.0.0.1'. The command entered is `LibraryDB> db.library.find({status:"Reserved"},{title:1,author:1,status:1})`. The output displays four documents from the 'library' collection, each with the fields '_id', 'title', 'author', and 'status'. The documents are for 'Harry Potter', 'The Jungle Book', 'Cinderella', and 'Panchatantra', all with a status of 'Reserved'.

```
LibraryDB> db.library.find({status:"Reserved"},{title:1,author:1,status:1})
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145b'),
    title: 'Cinderella',
    author: 'Charles Perrault',
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145c'),
    title: 'Panchatantra',
    status: 'Reserved'
  }
]
```

4.Fetch All Fields Except Category

Command

```
db.library.find({}, {category:0})
```



The screenshot shows a terminal window titled 'mongosh mongodb://127.0.0.1'. The command 'LibraryDB> db.library.find({}, {category:0})' has been entered. The output is a JSON array of three documents, each with fields: '_id', 'title', 'author', 'price', and 'status'. The 'category' field is excluded from the results. The documents are for 'Harry Potter', 'The Jungle Book', and 'Cinderella'. The terminal window is overlaid on a Windows desktop with a taskbar at the bottom showing various application icons and system status information like '33°C Mostly sunny' and '01:21 PM 03-10-2026'.

```
LibraryDB> db.library.find({}, {category:0})
[
  {
    _id: ObjectId('6ac09b841204f869914f1459'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    price: 500,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145a'),
    title: 'The Jungle Book',
    author: 'Rudyard Kipling',
    price: 300,
    status: 'Reserved'
  },
  {
    _id: ObjectId('6ac09de51204f869914f145b'),
    title: 'Cinderella',
    author: 'Charles Perrault',
    price: 400,
    status: 'Reserved'
  },
]
```